

CRISP-DM (Cross-Industry Standard Process for Data Mining) is the most widely used and longest-standing process framework for executing data mining, data science, and analytics projects. Developed in the late 1990s by a consortium including companies like SPSS, DaimlerChrysler, and NCR, it was officially published in 1999–2000 and remains the de-facto industry standard even today (still the most popular methodology according to multiple surveys up to recent years).

CRISP-DM is **non-proprietary**, **industry-independent**, and designed to be flexible. It describes data projects as an **iterative, cyclical process** rather than a strict linear sequence — you frequently loop back to earlier phases as new insights emerge.

The 6 Phases of CRISP-DM

Here is the classic structure (most sources describe it in this order):

1. Business Understanding

- Define the business problem and project objectives
- Understand success criteria (business + data mining)
- Assess situation, define data mining goals, and create a project plan
→ Goal: Make sure everyone agrees on **what** we are trying to solve and **how** success will be measured.

2. Data Understanding

- Collect initial data
- Describe, explore, and verify data quality
- Identify interesting subsets, patterns, anomalies, and first hypotheses
→ Goal: Get intimately familiar with the data you have (and what you still need).

3. Data Preparation

- Clean data (missing values, outliers, errors)
- Construct new attributes / features
- Integrate multiple data sources
- Format / transform data for modeling
→ Often the most time-consuming phase (frequently 60–80% of project effort).

4. Modeling

- Select appropriate modeling techniques
- Build and tune models (training, hyperparameters, etc.)
- Generate test designs
→ Goal: Create predictive / descriptive models that address the business question.

5. Evaluation

- Evaluate models from technical perspective (accuracy, precision, recall, AUC, etc.)
- Evaluate from business perspective (does it solve the real problem?)
- Review the entire process and decide next steps
 - Critical phase: often triggers loops back to previous phases.

6. Deployment

- Plan deployment (integration into business processes, reports, dashboards, APIs, etc.)
- Produce final reports and summaries
- Monitor & maintain the model in production
- Document everything for future reuse
 - Goal: Make the results actually useful in the real world.

The CRISP-DM process, including the six key phases and the important relationships between them (adapted from Wirth and Hipp, 2000, repr. in Kelleher *et al.*, 2020, p.14)

